

PROOF OF CONCEPT · DELIVERED

The S&P 500, as a graph that remembers time.

Domyn's FinReflectKG — a decade of S&P 500 **10-K** filings distilled into a knowledge graph — loaded into **ArangoDB** and made queryable in English, analyzable at scale, and reconstructable at any point in the last ten years.

17.5M

RELATIONSHIP FACTS

3.1M

DISTINCT ENTITIES

743

S&P 500 COMPANIES

2014–24

FISCAL YEARS

A dense, decade-long record of corporate self-disclosure.

FinReflectKG is distilled from the annual 10-K filings of the S&P 500 (research paper `2508.17906`). Every fact is grounded in the sentence that produced it.

- ◆ **17.5M relationship occurrences** across **3.1M** entities — 1.67 GB of raw parquet.
- ◆ **1.38M source text chunks** keep every edge traceable back to the filing.
- ◆ Nine entity kinds: `FIN_METRIC`, `ORG`, `GPE`, `RISK_FACTOR`, `PRODUCT`, `SEGMENT`, `FIN_INST`, `FIN_MARKET`, `PERSON`.

THE CHALLENGE

Metric-dominated & noisy. Financial metrics swamp centrality, OCR-era dates parse to absurd years, and anonymous mentions ("a supplier", "default") collapse into cross-company super-hubs. Turning this into a *trustworthy* graph was as much of the work as loading it.

One node collection, one edge collection, entity resolution for free.

A Labeled Property Graph in Neo4j terms: identity is `hash(name | type)`, so the same concept mentioned in 743 filings resolves to a single node automatically.



EDGES CARRY CONTEXT

`ticker`, `year`, `startDate` / `endDate`, `source` `chunk` — and the endpoint types, so a typed hop needs no vertex lookup.

SUPERNODE-SAFE

Two **vertex-centric indexes** turn typed 1-hops into direct edge queries. `net income` (in-degree $\sim 10^5$) scans only its **59,315** matching edges — not the whole hub.

The same graph, placed three ways — because placement is the performance story.

ArangoDB lets the *physical* layout vary while the model stays fixed. We built and benchmarked three (goal G7).

BASELINE

FinReflectKG

Flexible single-shard. The reference point for every comparison.

RECOMMENDED

OneShard

All collections co-located on one DBServer. **~4× faster on 2-hop** traversals — it removes the cross-DBServer `RemoteNode` hops.

LOCALITY PLAY

SmartGraph

Per-company concept duplication → **6.66M** nodes (2.1×), trading storage for company-local traversals.

Shared-cluster latency is noisy (203 ms–20 s for one query), so the **deterministic metrics are scanned-edge counts and shard-locality from `explain`** — not wall-clock.

A natural-language front door, with citations back to the filing.

English question → **Cypher** (via the `arango-cypher-py` integration, a first-class Cypher surface over ArangoDB) → executed → **GraphRAG** synthesis that grounds the answer in the retrieved edges and their source 10-K text.

- ◆ Every answer is **traceable** — edge provenance + chunk text, no free-floating claims.
- ◆ Cypher, not a bespoke DSL: the query layer is portable and inspectable.

HONEST EVALUATION

4 / 5

GraphRAG answer synthesis (gpt-4o). NL→Cypher: **19/22** transpile, **9/22** execute. LLM-dependent scores are provider-dependent and re-measured on provider change (§4.6).

Whole-graph algorithms, and an agent that plans them.

Pregel is deprecated on 3.12, so analytics run on the **Graph Analytics Engine** — PageRank and connected components over all 17.5M edges.

DETERMINISTIC BASE

PageRank + WCC verified on the full graph — the influence and community backbone the rest of the analysis builds on.

AGENTIC PLANNING

A pipeline turns a natural-language goal into an analysis plan, runs it on GAE, and writes a report — validated across **10 use cases**.

The **report stage is wired** — the tool's own generator renders Plotly HTML from stored GAE results (`scripts/analytics_report.py`). Remaining (optional, REQ-011): closing the loop into **fully-autonomous** NL → execute → report.

Ten years of filings means the graph can be rewound to any instant.

A valid-time interval model — adapted from our network-asset "TTL" blueprint, but as a *permanent archive* (no deletion). Validity lives **directly on the existing edges**; no overlay collection.

ENCODING

`validFrom` / `validTo` as `YYYYMM` ints, exclusive end, `NEVER_EXPIRES` sentinel for open rows.

QUERIES

as-of `vFrom ≤ t AND vTo > t` · **current** `vTo == NEVER` · **diff** $t_1 \rightarrow t_2 = \text{two as-ofs} + \text{MINUS}$.

INDEX-BACKED

An **MDI** index makes direct as-of scans index-covered; node-anchored typed as-of uses a composite VCI. Verified via `explain`.



When a fact was **asserted** vs. when it was **true**.

The filing `year` is the transaction-time axis; `validFrom` / `validTo` is the valid-time axis. Together they make the graph **bitemporal** — enabling "as-known-as-of" queries and restatement detection.

- ◆ **1.82M edges (10.5%)** are backward-looking — a filing asserting a fact about a period years earlier.
- ◆ Led by **utilities and insurers**, whose disclosures reach furthest back.

WHY IT MATTERS

You can reconstruct not just *what was true in 2018*, but *what the company said in 2018 about 2015* — the raw material for spotting restatements and revised guidance, with zero schema changes.

GAE PageRank on comparable, cleaned per-year snapshots. `net income` is #1 in every year — but the shape of what's central shifts.

2014 · TOP 8

ALL FINANCIAL METRICS

1	net income	FIN_METRIC
2	revenue	FIN_METRIC
3	net sale	FIN_METRIC
4	operate income	FIN_METRIC
5	earnings from operation	FIN_METRIC
6	fair value	FIN_METRIC

2024 · TOP 8

A VENUE BREAKS IN

1	net income	FIN_METRIC
2	revenue	FIN_METRIC
3	net sale	FIN_METRIC
4	new york stock exchange	▲ FIN_MARKET
5	operate income	FIN_METRIC
6	effective tax rate	FIN_METRIC

"CINC depends on Supplier" is a blank node, not a fact.

Extraction collapsed anonymous mentions — *a supplier, a default* — into single cross-company super-hubs. This is exactly RDF's blank-node problem, and it was poisoning centrality.

▸ RAW — CONFLATED HUBS

supplier distributor competitor default other

One "supplier" node wired to hundreds of unrelated companies — a phantom super-hub that outranked real concepts.

▾ CLEANED — SKOLEMIZED

bn_aapl_supplier bn_cinc_supplier bn_msft_vendor

5,893 per-company blank nodes minted; 23 junk placeholders excluded (not skolemized). 89 hubs flagged in all.

Discriminator is **role-lexicon × cross-company fan-in**, not fan-in alone — so genuinely shared concepts (common stock: 716 companies, dividend, long-term debt) stay intact. PageRank was re-run on the cleaned snapshots.

Delivered end-to-end — and set up to go further.

▼ SHIPPED

- ◆ Full load + **three sharded distributions** benchmarked (G5/G7)
- ◆ NL→Cypher + **GraphRAG** with citations (G6)
- ◆ GAE analytics + **agentic planning**, 10 use cases (G8)
- ◆ **Time-travel + bitemporal** + per-year PageRank (G9)
- ◆ Generic-mention cleanup + **live demo** (G9-P5)

→ NEXT

- ◆ Fold `validFrom` / `validTo` into shared ETL so **all three distributions** inherit time-travel — then benchmark it (the G7 lens)
- ◆ Close the **fully-autonomous** NL→execute→report loop (REQ-011) — the report stage now exists; what remains is the agent driving it end to end
- ◆ Capture a time-slider **animation** for the deck